

MSCR Tracker — Performance Analytics

Session: 319 frames | 14.4 s | Resolution 848×480 @ 30 fps

0.9 %

3-D Detection
Rate

16.3 %

Skeleton
Found Rate

37.6 ms

Mean Proc.
Time

26.6 fps

Throughput
(est.)

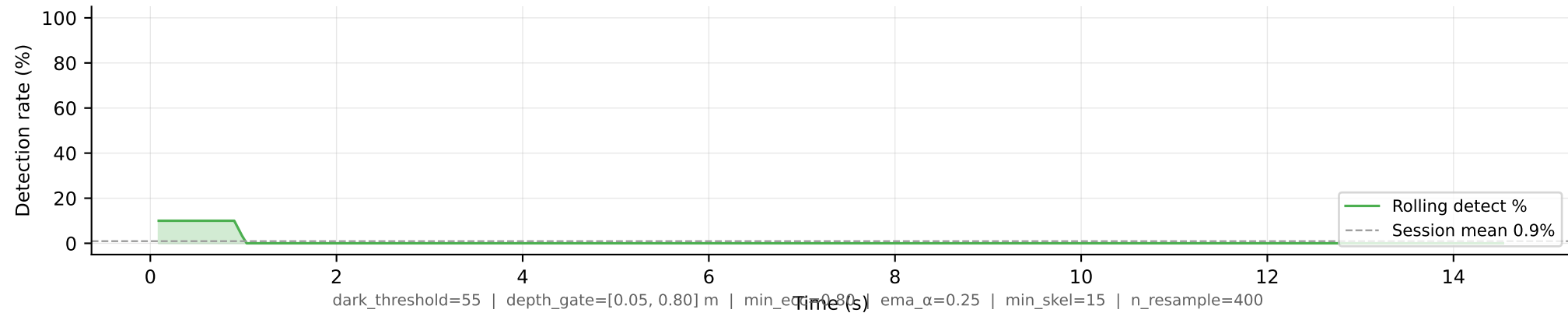
41.2 mm

Mean Arc
Length

499.75 mm

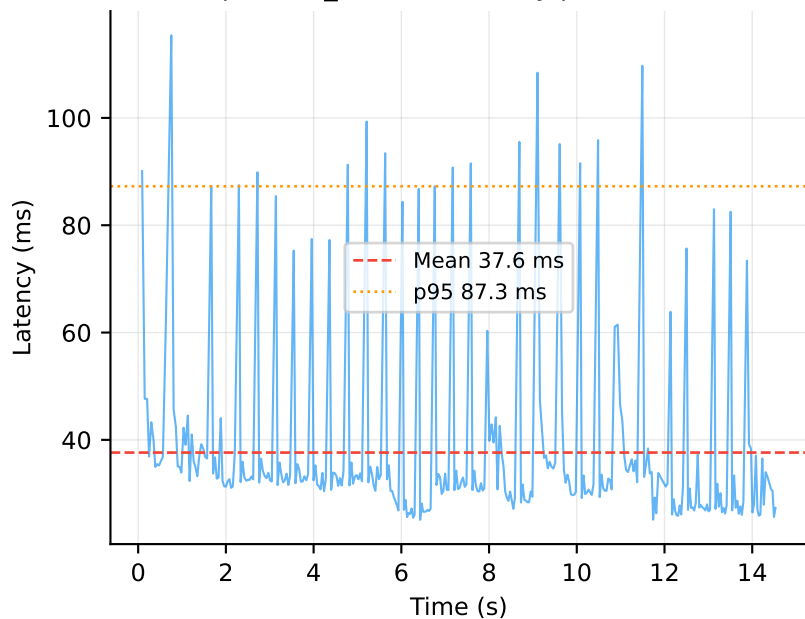
Tip Jitter
(p95)

Rolling 3-D Detection Rate (30-frame window)

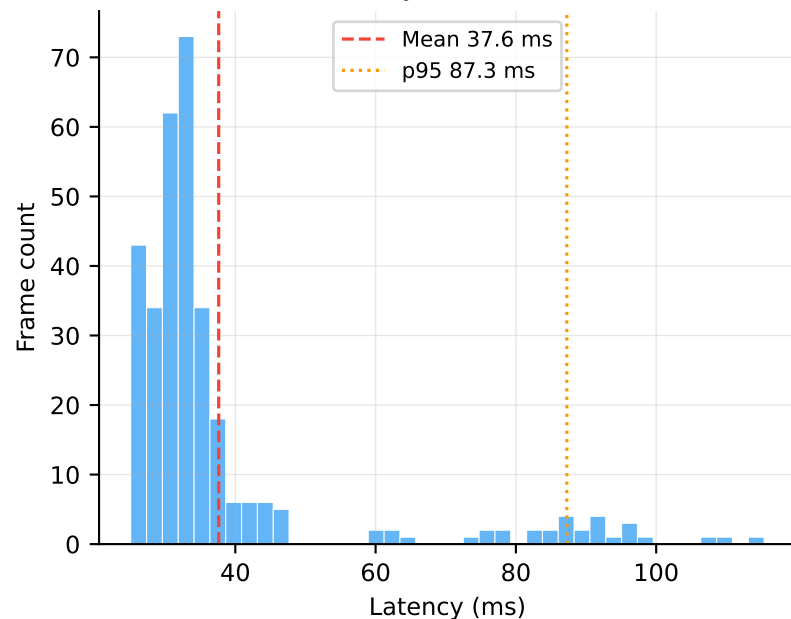


Processing Latency & Skeleton Quality

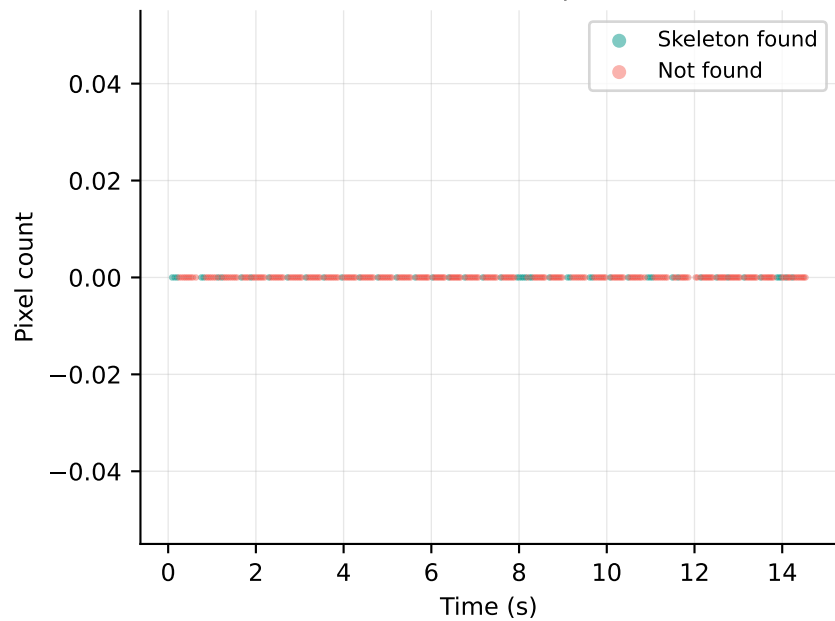
process_frame() Latency per Frame



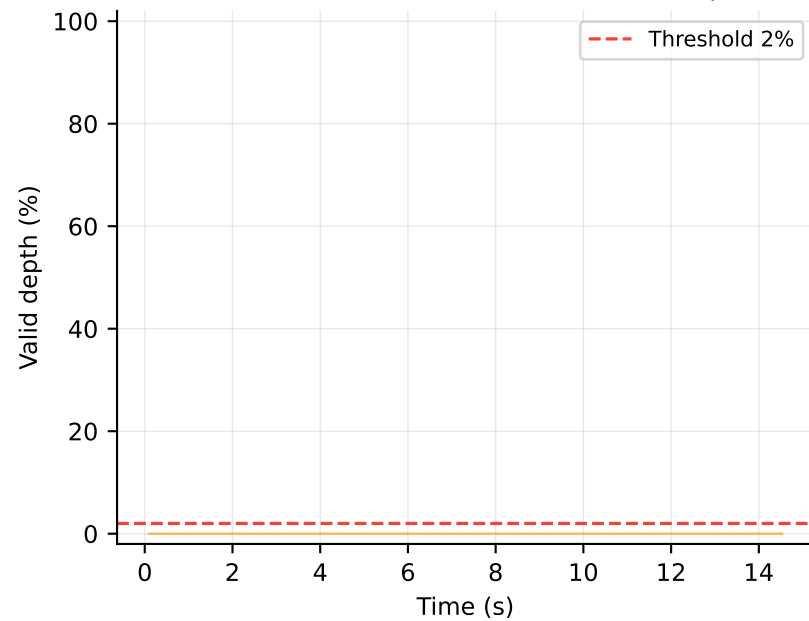
Latency Distribution



Skeleton Pixel Count per Frame

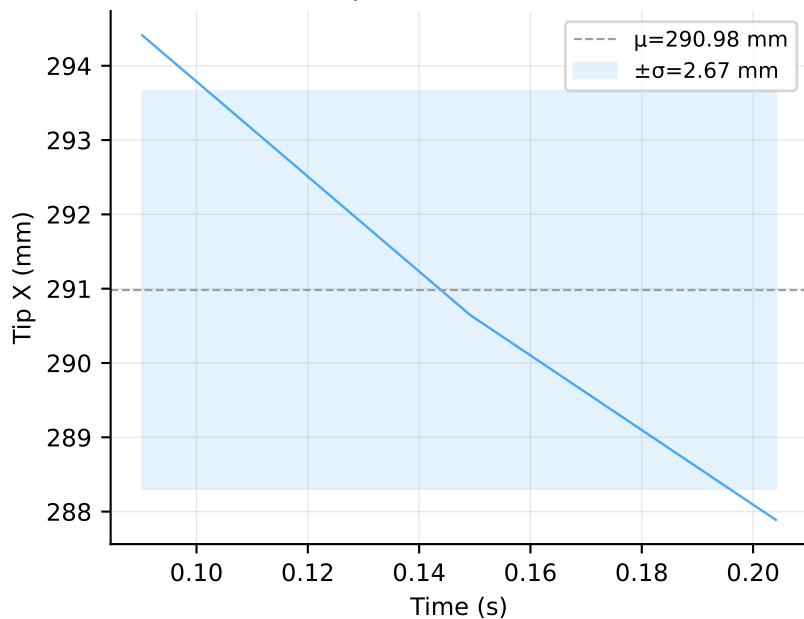


Fraction of Skeleton Pixels with Valid Depth

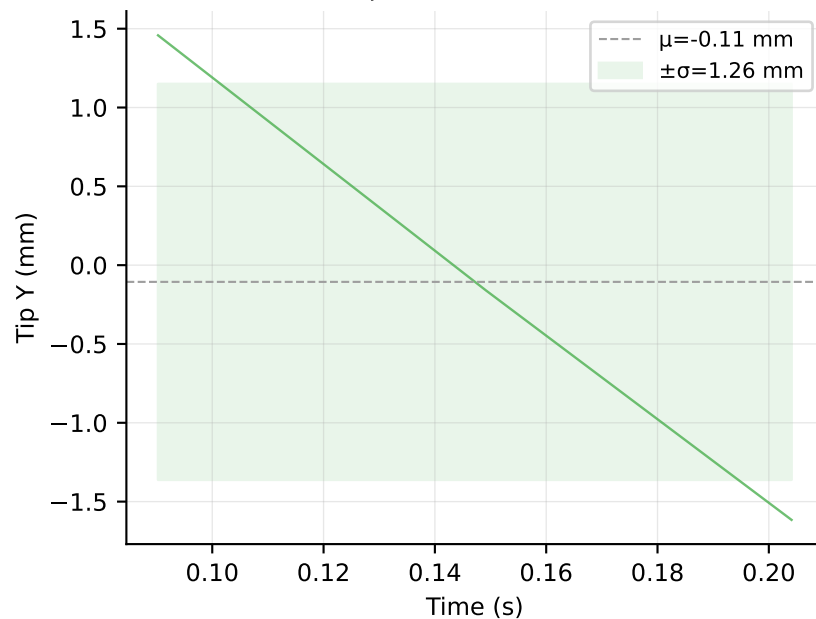


3-D Tip Position Trajectories

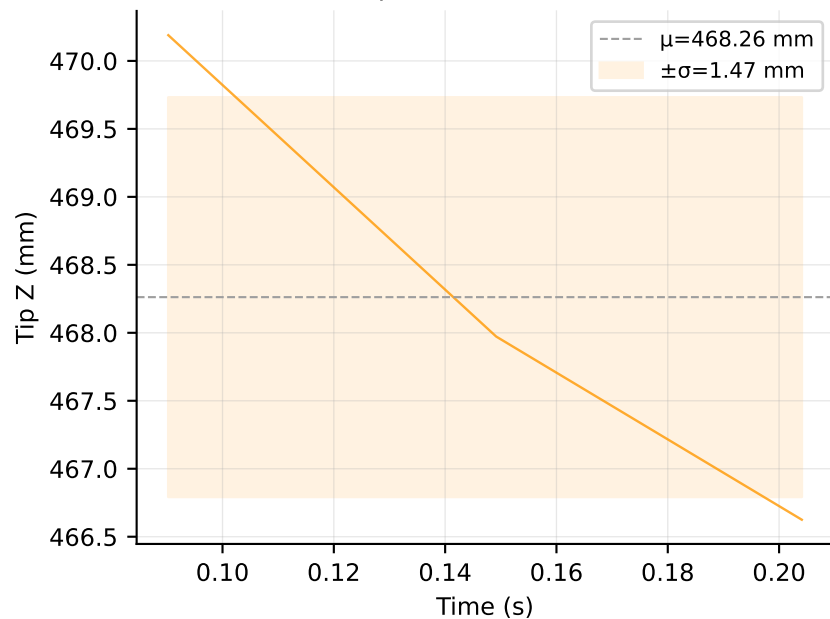
Tip X (mm) vs Time



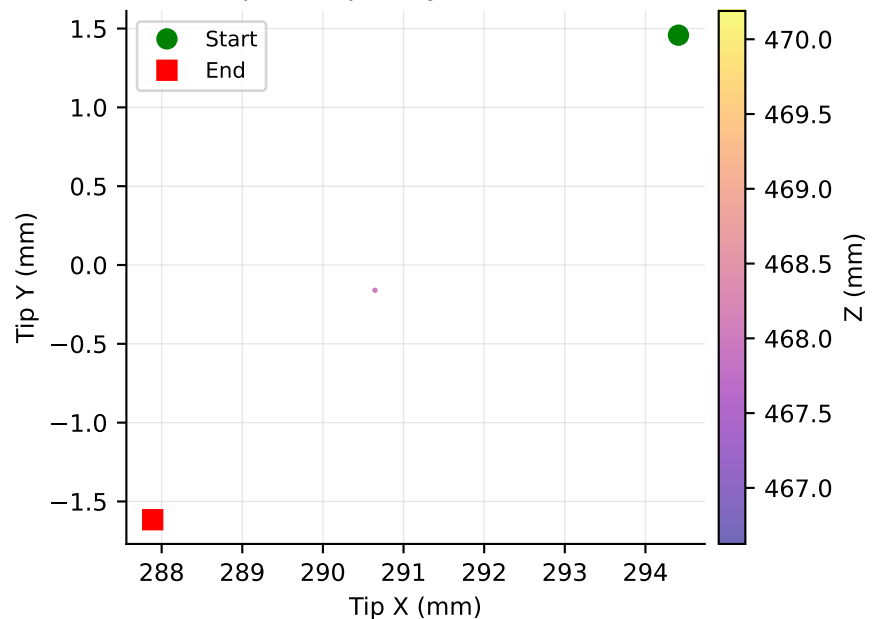
Tip Y (mm) vs Time



Tip Z (mm) vs Time

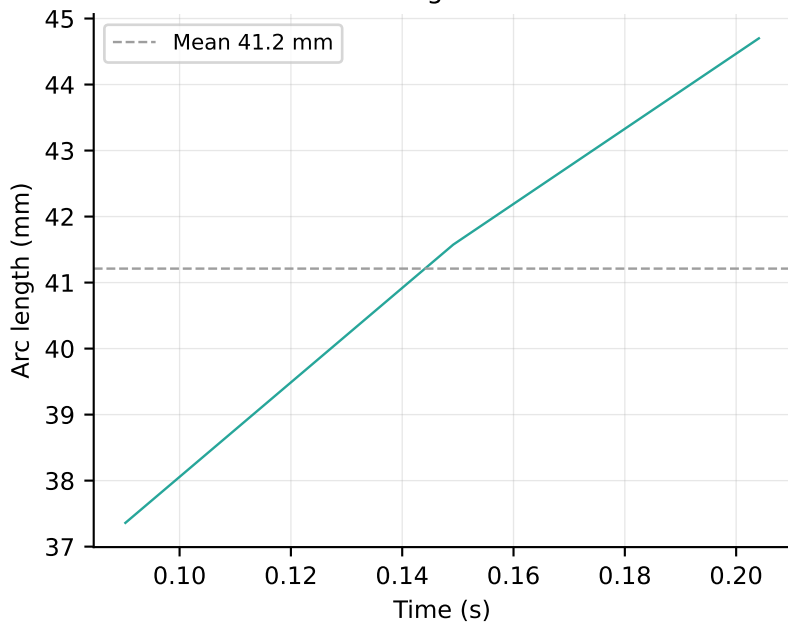


Tip XY Trajectory (colour = Z)

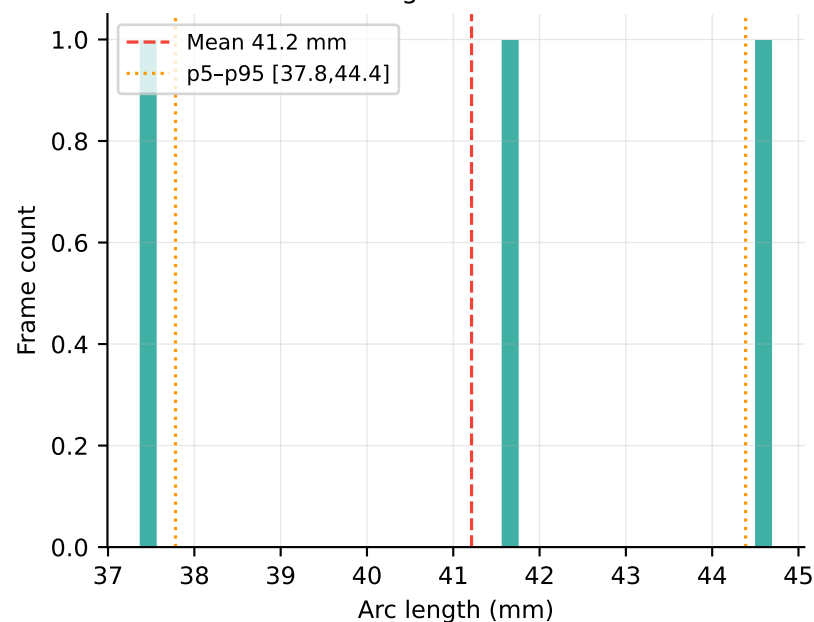


Arc Length & Tip Stability

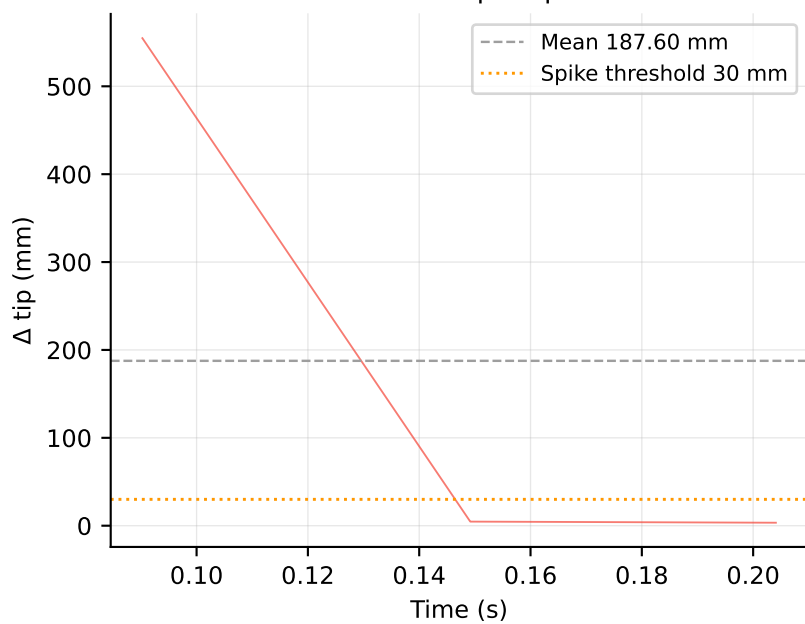
3-D Arc Length over Time



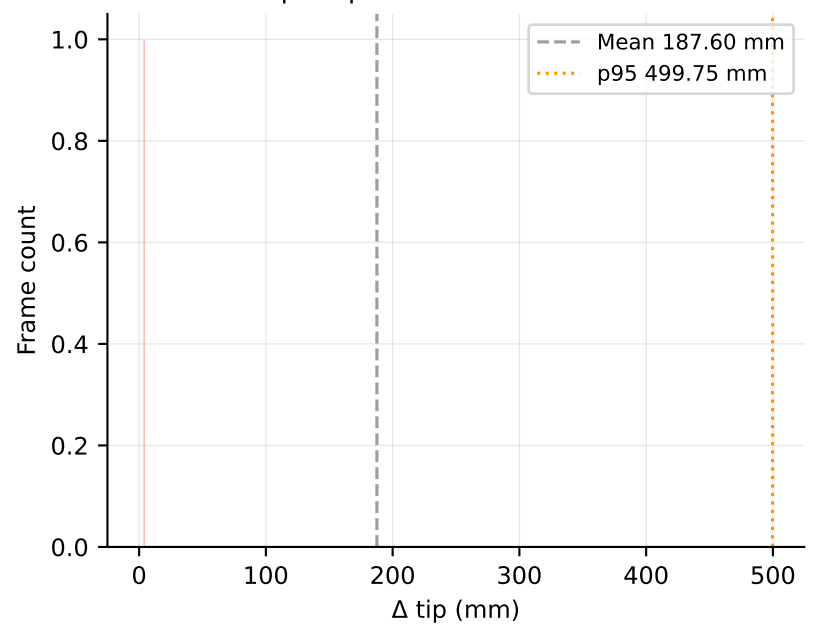
Arc Length Distribution



Frame-to-Frame Tip Displacement



Tip Displacement Distribution



Summary Statistics Table

Metric	Mean	Std	Min	Max	p5	p95	Unit
Process latency	37.632	17.045	25.089	115.389	26.311	87.259	ms
Skeleton pixel count	0.000	0.000	0.000	0.000	0.000	0.000	px
Valid depth fraction	0.000	0.000	0.000	0.000	0.000	0.000	%
Arc length	41.212	3.008	37.360	44.700	37.781	44.387	mm
Tip X	290.981	2.673	287.889	294.410	288.164	294.033	mm
Tip Y	-0.106	1.255	-1.616	1.458	-1.470	1.296	mm
Tip Z	468.262	1.469	466.625	470.189	466.760	469.968	mm
Tip jump Δ	187.604	259.618	3.396	554.759	3.523	499.749	mm

Total frames: 319 | 3-D valid: 3 (0.9%) | Skeleton found: 52 (16.3%) | Duration: 14.4 s | Mean throughput: 26.6 fps

Note: 'Tip jump' is frame-to-frame Euclidean distance of the smoothed tip in 3-D (mm). Only computed on consecutive valid frames.